Project Outline: H&M Material Sustainability Intelligence Prototype

This project applies data analytics and visualization techniques to a real sustainability problem in the fast-fashion industry. It illustrates how unstructured product metadata can be converted into actionable sustainability intelligence, aligning with business analytics competencies such as data transformation, scenario modeling, and decision support. The project aims to design and implement a data analytics prototype that transforms H&M's 2024 product material data into measurable sustainability insights, demonstrating how material-level analysis can support data-driven decision-making for sustainable fashion production and supply chains.

According to H&M's 2024 Annual and Sustainability Report, material production remains one of the largest contributors to the company's overall greenhouse gas (GHG) emissions, representing around 44% of total Scope 3 emissions, followed by the use of sold products and raw material extraction. This highlights the critical role of material composition in shaping H&M's carbon footprint and underlines the importance of developing analytical tools that can assess, monitor, and optimize material sustainability performance across product lines.

We have defined 4 objectives:

1. Develop a sustainability scoring model based on material composition and estimated environmental impact.
2. Aggregate and compare sustainability scores across product categories and material types.
3. Conduct a what-if simulation to estimate potential CO_2 reductions from increased use of sustainable materials.
4. Visualize results through an interactive dashboard prototype to support business and sustainability stakeholders.

In order to address this challenge, we will use the following public data: H&M Product Dataset (2024), that contains product ID, name, category, and detailed material composition (e.g., "60% recycled cotton, 40% polyester").

The analytical process will begin with data cleaning and parsing, where the product material information is standardized and transformed into a structured format suitable for analysis. Text-based material descriptions (e.g: "60% recycled cotton, 40% polyester") will be decomposed into identifiable components, enabling quantitative assessment. Each material will then be assigned an estimated CO_2 emission factor and a sustainability weight derived from life-cycle assessment (LCA) data. This will allow for the calculation of a sustainability index at the product level, reflecting both the environmental impact and the proportion of sustainable materials.

Subsequent analysis will aggregate the results by product category and material type to identify high-impact areas within H&M's product portfolio. Comparative analysis will evaluate differences between recycled and non-recycled materials, highlighting potential opportunities for improvement. Finally, a what-if simulation will model alternative material compositions. For instance, substituting polyester with recycled polyester, to estimate potential reductions in overall CO_2 emissions. This analytical framework provides a clear, data-driven foundation for assessing and optimizing material sustainability performance across H&M's product range.

Our expected outcomes, include:

- A Material Sustainability Index for H&M products.
- Visual dashboard showing material mix, CO_2 footprint, and sustainability distribution.
- Quantified insights on how material substitution impacts total emissions.
- Prototype demonstrating the potential of data analytics for sustainability governance in retail fashion.

GHG emission reduction targets[1,2,3]

	2024	2023	2022	2019 (Baseline)	Target (year and value)	% change compared with 2023	% change compared with baseline
Total scope 1 emissions (tonnes CO_2e)	15,102	15,754	18,048	21,564	2030, −56%	−4%	−30%
Total location based scope 2 emissions (tonnes CO_2e)	364,693	377,573	443,099	658,646	No target	−3%	−45%
Total market based scope 2 emissions (tonnes CO_2e)	26,553	39,508	46,806	48,735	2030, −56%	−33%	−46%
Total scope 1 and scope 2 market based emissions (tonnes CO_2e)	**41,655**	**55,262**	**64,854**	**70,299**	**2030, −56%**	**−25%**	**−41%**
Total scope 3 emissions (tonnes CO_2e) under our science based target (excluding use-phase emissions)	6,955,000	6,754,000	7,591,000	9,115,000	2030, −56%	3%	−24%
Total scope 3 emissions including use-phase emissions (tonnes CO_2e)	8,729,000	8,413,000	9,442,000	11,613,000	No target	4%	−25%
Total GHG Emissions (tonnes CO_2e)	**8,770,000**	**8,469,000**	**9,507,000**	**11,684,000**	**2030, −56%**	**4%**	**−25%**
Total scope 1 emissions intensity value (tonnes CO_2e/MSEK)	0.06	0.07	0.08	0.09	No target	−4%	−30%
Total market based scope 2 emissions intensity value (tonnes CO_2e/MSEK)	0.11	0.17	0.21	0.21	No target	−32%	−46%
Total scope 3 emissions intensity value (tonnes CO_2e/MSEK)	37.2	35.6	42.2	49.9	No target	4%	−25%
Total scope 3 emissions including use-phase emissions intensity value (tonnes CO_2e/MSEK)	29.7	28.6	34.0	39.2	No target	4%	−24%
Total GHG Emissions intensity value (tonnes CO_2e/MSEK)	**37.4**	**35.9**	**42.5**	**50.2**	**No target**	**4%**	**−25%**

1. In 2024, we have made some updates in how we calculate emissions compared with 2023, please see page 66 on the progress and updated methods for more details of these changes. Read more about our data and calculation methods at hmgroup.com/sustainability/sustainability-reporting/how-we-report.
2. Scope 1 emissions are all direct emissions from our own operations; scope 2 represents indirect GHG emissions from consumption of purchased electricity, heat or steam used in our own operations; scope 3 includes other indirect emissions, such as the extraction and production of purchased materials and fuels, transport-related activities in vehicles not owned or controlled by us, energy-related activities not covered in scope 2, outsourced activities, and waste disposal. Scope 1 and 2 limitations and comments: only stores open for the full quarter are included; electricity consumption includes both actual data and estimates, where estimates are made if actual data is not received or not available within the reporting deadline. Electricity consumption data for HVACs operated by landlords is estimated.
3. Scope 3 emissions data are rounded to the nearest thousand.

Scope 3 emission split by category[1,2]

Tonnes	Scope 3 category	2024	2023	2022	2019 (Baseline)	% change compared with 2023	% change compared with baseline
Raw materials[3]	1	1,180,000	1,131,000	1,222,000	1,849,000	4%	−36%
Fabric production[3]	1	3,872,000	3,757,000	4,280,000	4,826,000	3%	−20%
Garment manufacturing[3]	1	280,000	252,000	305,000	360,000	11%	−22%
Non-garment goods[3]	1	334,000	336,000	393,000	454,000	0%	−26%
Packaging	1	110,000	145,000	152,000	235,000	−24%	−53%
Other expenditures[3]	1	561,000	601,000	672,000	706,000	−7%	−20%
Transport[3]	4	405,000	306,000	335,000	453,000	32%	−11%
End-of-life sold products[3]	12	65,000	62,000	66,000	73,000	5%	−11%
Other							
Fuel & energy related emissions	3	28,000	25,000	31,000	33,000	13%	−25%
Waste in own operations	5	3,000	4,000	3,000	4,000	−25%	−18%
Business travel	6	26,000	18,000	8,000	23,000	47%	15%
Employee commuting	7	25,000	28,000	30,000	44,000	−10%	−43%
Franchise	14	32,000	50,000	59,000	50,000	−35%	−36%
Investments	15	34,000	40,000	35,000	6,000	−14%	434%
Not included in Science based targets							
Use of sold products	11	1,774,000	1,659,000	1,851,000	2,499,000	7%	−29%

1. For categories 2 capital goods, 8 & 13 leased assets, and 10 processing of sold products, no activities with relevant climate impact has been identified, therefore these are excluded from disclosure. We periodically evaluate if any relevant emissions occur from these sources.
2. Emissions data are rounded to the nearest thousand.
3. Emissions have been recalculated in the area since the 2023 report, please see page 66 on the progress and updated methods for more details of these changes.

Total GHG emissions (tonnes CO_2e)

The charts show scope 3 emissions split by category as share of total scope 3 emissions (on the right) and our progress in reducing scope 1, 2 and 3 emissions in the different areas of our value chain against a 2019 baseline (below).

Scope 3 data 2024